

KKK Oil Factory Enterprise ERP & Billing System

Comprehensive Path-Wise, Button-Wise, and Operational Manual for Web, Desktop, and Mobile Apps. Details all user roles, statutory GST workflows, expeller manufacturing, and supply chain controls.

SYSTEM EDITION	ENTERPRISE	SUPPORTED ROLES
v2.5.0 Standard GST	KKK Oil Factory, Tamil Nadu	Super Admin, Admin, Manager, Cashier, Supervisor, Driver
PLATFORMS		
Web (Chrome/Edge/Safari), Android APK, iOS		

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Mobile Application User Guide

The **KKK Oil Factory Enterprise ERP** provides complete end-to-end digitisation for edible oil extraction mills and commercial distribution. The application operates simultaneously on desktop workstations and handheld mobile devices using an adaptive responsive shell.

Desktop vs. Mobile Experience

- **Desktop Workstations ($\geq 860\text{px}$):** Features a permanent 244px dark sidebar with gold accents, top bar with global search and notification bells, and wide data tables for efficient keyboard-driven data entry.
- **Mobile Smart Devices ($< 860\text{px}$):** Provides a streamlined 5-station bottom navigation bar (`Dashboard` , `Billing` , `Inventory` , `History` , and `More Modules` bottom sheet) with touch-friendly buttons and mobile thermal printer integration.
- **Force Desktop View Toggle:** On tablet devices or mobile screens, tapping the topbar `Desktop View` button forces full desktop table mode, enabling viewing of complex multi-column billing sheets.

The software enforces strict privilege boundaries to protect company profit margins, customer financial ledgers, and raw seed stocks:

Role Key	Display Role	Tone	Access Level	Allowed Modules & Scope
super_admin	Super Admin	Gold	Full System (*)	Master pricing, company settings, employee administration, GST & Non-GST stations, database export.
admin	ERP Administrator	Gold	15 Core Modules	Master product catalog, customer accounts, suppliers, inventory adjustments, daybook reports.
manager	Plant / Shift Manager	Blue	12 Modules	Cashier shift monitoring, production yield approval, purchase orders, customer statements, logistics dispatches.
cashier	Billing Cashier / POS	Green	Billing & POS	Fast POS counter sales, B2B tax invoicing, payment collection, thermal receipt printing, customer balance viewing.
production_staff	Production Supervisor	Amber	Factory Floor	Expeller batch logging, raw seed crushing input, oil output recording, oil cake bagging, machine hours.
logistics_staff	Logistics Driver	Purple	Fleet & Dispatch	Gate pass generation, delivery run-sheets, trip status tracking, customer Proof of Delivery (POD).
inventory_incharge	Warehouse Head	Forest	Stock & Tanks	Bulk oil silo dip stick measurements, packaged goods stock, packaging materials (tins/bottles), stock transfers.

Path: /login • Route: login

Public / All Personnel

Authenticates company personnel, verifies cryptographically hashed passwords, creates session tokens in local storage, and loads permissions.

Form Input Fields

- **Username** : Staff account handle (e.g. `admin` , `admin_staff` , `mgr` , `cashier`).
- **Password** : Password field. Includes an eye toggle icon to switch between concealed dots and plain-text.

Action Buttons

- **Sign In to ERP System** : Authenticates user, initializes reactive state via `AuthProvider` , and navigates to the authorized start screen.
- **Quick Demo Buttons** : Single-click pre-fills credentials for `admin` , `admin_staff` , `mgr` , and `cashier` .

Path: /dashboard (or /) · Route: dashboard

Super Admin, Admin, Manager

Real-time operational dashboard providing instant visibility into daily mill throughput, cash balances, silo tank levels, and inventory alerts.

Metric KPI Cards

- **Today's Gross Sales:** Total rupees billed across all GST tax invoices and counter sales.
- **Cash Register Balance:** Physical cash collected and on hand in cashier drawers.
- **Bulk Silo Tank Levels:** Live volume of oil stored in main groundnut and sesame storage tanks.
- **Low Stock Alerts:** Count of packaged goods whose inventory has fallen below safety minimum.

Quick Action Buttons

- **+ New B2B Invoice** : Fast-tracks user to `/billing` for wholesale tax invoicing.
- **+ Quick POS Counter** : Opens the high-velocity retail sales screen at `/counter` .
- **+ Start Milling Batch** : Opens the batch recording modal for expeller runs.
- **+ Record Seed Inward** : Launches the weighbridge procurement dialog.
- **+ Delivery Gate Pass** : Opens logistics dispatch tracking at `/logistics` .

Path: /billing (Aliases: /gst-billing, /erp-billing) • Route: erp_billing

Super Admin, Admin, Manager, Cashier

The central enterprise billing engine for wholesale orders, dealers, distributors, supermarkets, and institutions. Formats tax invoices strictly conforming to **Rule 46 of CGST Rules, 2017**.

Header Form Controls

- Customer Selector** : Searchable dropdown displaying customer legal name, GSTIN, and current balance.
- Route** : Target delivery corridor (e.g. Dharmapuri Local, Salem Highway, Coimbatore Express).
- Godown Selector** : Deducts stock from **Main Godown** , **Packaging Godown** , or **Mill Floor** .
- Pricing Tier** : Radio buttons for **Retail Rate** , **Semi-Wholesale Rate** , or **Bulk Wholesale Rate** .
- Payment Mode** : Dropdown options: **Cash** , **Credit / Account** , **UPI / QR** , **Bank Transfer** , **Cheque** .
- Payment Status** : **Paid** , **Partial** , or **Credit (Unpaid)** .

Line Item Entry & Automation

- Item Search Autocomplete** : Type SKU code or name; automatically fetches HSN, packaging unit, and rate.
- Quantity** : Number of units to add.
- Applied Unit Rate** : Auto-populated from chosen pricing tier; editable by authorized managers.
- Discount %** : Line discount percentage (e.g., 2.0%).
- + Add to Voucher** : Inserts product into line items table with automatic tax calculations.

Automated GST Tax Engine

The system compares the customer's 2-digit GST state code with the factory code (Tamil Nadu: 33):

- Intra-State Sale (Same State: 33)**: Applies **CGST 2.5%** + **SGST 2.5%** = **5.0% Total Tax**.
- Inter-State Sale (Out of State: e.g. 29, 32)**: Applies **IGST 5.0%** .

Bottom Multi-Tab Panel

- **Tab 0 (Dispatch Details):** Input Buyer PO Number, PO Date, Vehicle Number (e.g. `TN 28 BK 5521`), Driver Name, and Delivery Challan No.
- **Tab 1 (E-Way Bill):** Mandatory 12-digit E-Way Bill Number and validity duration for shipments over ₹50,000.
- **Tab 2 (E-Invoice):** 64-character IRN hash, QR code data, and Government portal acknowledgement.
- **Tab 3 (Tax Ledgers):** Comprehensive ledger breakdown showing pre-tax subtotal, CGST, SGST, IGST, round-off, and grand total.

Action Buttons

- **Save Invoice** : Validates all inputs, deducts godown stock, updates customer credit ledger, creates permanent voucher ID (`V-2026-XXXX`), and saves to Firestore.
- **Print A4 Tax Invoice** : Generates statutory 300 DPI A4 PDF invoice complete with bank details and signature block.
- **Print Thermal Receipt** : Emits 80mm / 58mm ESC/POS slip.
- **Clear Form** : Resets all fields for next bill.

Path: /billing-history (Alias: /bills) · Route: billing_history

Super Admin, Admin, Manager, Cashier

Chronological audit repository of all issued GST tax invoices. Allows searching, reprinting, and inventory-restoring cancellations.

Filters & Search

- **Search Invoices** : Live search by voucher number (e.g. `V-2026-001`), customer name, or billing date.
- **Payment Status Chips** : Filter table by `All` , `Paid` , `Credit` , or `Partial` .

Table Action Buttons

- **Reprint A4** : Re-opens PDF invoice preview with 1-click download and print.
- **Reprint Thermal** : Sends invoice copy to thermal receipt printer.
- **Cancel Bill** :
 - Prompts for staff cancellation reason.
 - Immediately restores all product quantities back into warehouse stock.
 - Reverses the customer's outstanding balance debit.
 - Marks voucher as `Cancelled` in tax audit logs.

Path: /counter (Alias: /pos) • Route: counter

Cashier, Manager, Admin

High-velocity touch screen cash register designed for factory outlet retail counters. Provides rapid product selection, instant cart steppers, and automated change return calculation.

Product Catalog Grid & Filter Chips

- Filter catalog by `All`, `Groundnut Oil`, `Sesame Oil`, `Coconut Oil`, or `Mustard Oil`.
- One-tap on any product card adds 1 unit to the cart.
- Price Tier Chip switcher: `Retail`, `Semi-Wholesale`, `Bulk Wholesale`.

Cart Controls & Cash Tender Calculator

- `+ / - Steppers` : Rapid quantity adjustments per cart line.
- `Discount Field` : Custom promotional discount percentage.
- `Payment Mode Chips` : `Cash`, `UPI QR`, `Card`, `Credit`.
- `Tendered Cash Input` : Cashier enters cash handed by customer; system automatically computes **Change to Return (₹)**.

Action Buttons

- `Complete & Print Thermal` : Finalizes POS checkout, deducts stock, and prints thermal slip.
- `Complete & Print A4` : Completes transaction with A4 invoice.
- `Hold Cart` : Temporarily holds customer cart in memory.
- `Clear Cart` : Wipes active cart.

Path: /products · Route: products

Super Admin, Admin, Manager

Central master registry of all edible oils, packaged SKUs, and byproduct oil cakes. Manages tier pricing (AWR architecture), HSN codes, and safety reorder levels.

Header Buttons

- **+ Add New Product Master** : Opens comprehensive SKU creation modal.
- **Export Master CSV** : Exports entire catalog to spreadsheet.

Add / Edit Product Modal Details

- **Product Name** : Formal name (e.g. *Cold Pressed Groundnut Oil 15L Tin*).
- **Product Code & SKU** : Internal identifier (e.g. PRD-GN-15L) and barcode.
- **Brand & Oil Type** : Brand name and seed type (Groundnut, Sesame, Coconut).
- **Packaging Unit** : 15L Tin, 5L Can, 1L Bottle, 500ml Pouch, 50kg Bag.
- **Tiered Rates (AWR)** :
 - **Agency Rate (₹)**: Base price for wholesale dealers and distributors.
 - **Wholesale Rate (₹)**: Standard rate for semi-wholesale stores.
 - **Retail Rate (₹)**: Counter price for walk-in retail buyers.
 - **MRP (₹)**: Maximum Retail Price printed on consumer pack.
- **Stock & Thresholds** : Physical stock on hand and minimum safety alert level.
- **Tax & HSN** : GST Rate (5%) and HSN code (1508 / 1515 / 2306).

Path: /price-management · Route: price_management

Super Admin, Admin

Dynamic commodity pricing engine that protects mill profit margins against daily agricultural seed market fluctuations.

Margin Protection Formula

$$\$ \text{ ext{Gross Margin (₹/L)} } = \text{ ext{Selling Rate} } - (\text{ ext{Seed Cost} } + \text{ ext{Milling Cost} } + \text{ ext{Packaging} } - \text{ ext{Cake Recovery} }) \$$$

Action Buttons

- Bulk % Price Revision** : Adjusts an entire product category up or down by a set percentage.
- Save Price Changes** : Commits revised rates to database and pushes updates to all billing terminals.

Path: /price-history · Route: price_history

Super Admin, Admin, Manager

Historical analytics comparing regional mandi raw seed arrival prices against finished packaged oil wholesale rates. Features audit logs detailing which administrator modified product prices.

Path: `/inventory` (Alias: `/stock`) · Route: `inventory`

Super Admin, Admin, Manager, Warehouse Head

Dual-mode inventory tracking covering both bulk oil silos (via physical dip stick calibration) and packaged finished goods.

Bulk Silo Tanks (Dip Calibration)

- Monitors Tank 1 (Raw Groundnut Oil), Tank 2 (Sesame Oil), Tank 3 (Coconut Oil), and Tank 4 (Polished Finished Oil).
- Displays calibrated dip stick height in cm, converted volume in Liters, and capacity percentage.

Action Buttons

- **Adjust Physical Stock** : Opens dialog to record physical stock counts and reconcile discrepancies.
- **Stock Transfer Note** : Records internal transfer between factory warehouse and retail counters.
- **Delete Inventory Item** : Deletes obsolete item from inventory records.

Path: /purchase (Alias: /inward) • Route: purchase

Super Admin, Admin, Manager

Manages raw oil seed procurement from agricultural mandis and farmer suppliers with weighbridge gross/tare tracking and QC grading.

Header Action Button

- **+ Record Seed & Raw Material Inward** : Opens Goods Receipt Note (GRN) dialog.

GRN Intake Modal Details

- **Material Name** : Groundnut Raw Pods, Sesame Seeds, Copra, Mustard Seeds.
- **Supplier Dropdown** : Selects vendor from supplier master directory.
- **Quantity (kg)** : Net weight from weighbridge ticket.
- **Rate per kg (₹)** : Agreed procurement rate per kg.
- **QC Status** : **Passed (Graded)** (ready for crushing) or **Pending Lab Test** (quarantine for moisture test).
- **Save Inward Entry** : Saves procurement record and updates vendor payable ledger.

Path: /production (Alias: /milling) • Route: production

Super Admin, Admin, Manager, Production Supervisor

Monitors expeller crushing runs, calculating extraction recovery yields, byproduct cake generation, and processing moisture loss.

Header Action Button

- **+ Start New Production Milling Run** : Opens crushing run logging dialog.

Crushing Batch Dialog Fields

- **Target Product** : Finished oil SKU being manufactured.
- **Planned Seed Input (kg)** : Total raw seeds fed into expellers.
- **Filtered Oil Output (Liters)** : Recovered polished oil pumped into storage tanks.
- **Oil Cake Output (kg)** : Total cattle feed byproduct cake bagged.
- **Wastage (Liters)** : Processing loss and sediment.
- **Total Seed Cost (₹)** : Valuation of crushed seeds.
- **Record Milling Run** : Commits batch record (**PRD-BXXXX**), computes yield percentage, and increases bulk oil inventory.

Path: /customers • Route: customers

Super Admin, Admin, Manager, Cashier

Comprehensive customer registry tracking retail stores, wholesale dealers, GSTIN numbers, delivery routes, and credit limits.

Customer Segmentation

- **Regular / Retail** : Small retail groceries, supermarkets, and cash buyers.
- **Dealer / Wholesaler** : Regional distributors with high-volume price tiers and credit terms.

Action Buttons

- **+ Add New Customer Account** : Opens customer registration modal.
- **Edit Customer** : Updates address, phone, GSTIN, or credit limit.
- **View Statement / Ledger** : Generates chronological account statement showing invoice debits and payment credits.
- **Delete** : Removes customer profile if no unpaid invoices exist.

Path: /suppliers · Route: suppliers

Super Admin, Admin, Manager

Directory of agricultural mandi brokers, farmer cooperatives, tin manufacturers, and packaging suppliers. Tracks outstanding mill payables.

Action Buttons

- **+ Add New Supplier Vendor** : Opens vendor registration modal.
- **Edit Supplier** : Updates contact info, materials, and GSTIN.
- **Record Payment Voucher** : Records disbursement of funds against outstanding payable dues.
- **Delete** : Removes vendor profile.

Path: /logistics (Alias: /dispatch) • Route: logistics

Super Admin, Admin, Manager, Logistics Driver

Vehicle loading verification, gate pass issuance, driver trip sheets, and customer Proof of Delivery (POD) mobile confirmation.

Action Buttons

- **+ Generate Dispatch Gate Pass** : Opens modal to link Vehicle Number, Driver, Route, and Sales Invoice.
- **Issue Gate Pass** : Commits gate pass (DSP-XXXX) and sets status to **Dispatched** .
- **Status Stepper / POD** : Driver marks delivery as **In Transit** and toggles **POD Received** on mobile upon delivery.

Path: /reports (Alias: /analytics) · Route: reports

Super Admin, Admin, Manager

Financial analytics, cashier daybook cash reconciliation, and statutory GSTR-1 tax compliance summaries.

Key Tax Summaries

- **B2B Registered Supplies (GSTR-1 Table 4):** Invoices issued to customers with valid GSTIN numbers.
- **B2C Consumer Supplies (GSTR-1 Table 7):** Walk-in counter retail sales without buyer GSTIN.
- **Tax Liability Split:** CGST (2.5%), SGST (2.5%), and IGST (5.0%).
- **HSN Summary Table (Table 12):** Itemized totals by HSN code (1508 , 1515 , 2306).

Action Buttons

- **Export GSTR-1 Excel / CSV** : Downloads portal-compatible tax spreadsheet for monthly filing.
- **Download Daily Daybook PDF** : Generates comprehensive daily cash ledger report.

Path: /employees (Alias: /staff) • Route: employees

Super Admin, Admin

User account administration, staff password provisioning, and the 15-module fine-grained permissions matrix.

Staff Controls & Buttons

- **+ Add New Employee** : Opens modal to create username, password, phone, email, and role.
- **Password Eye Icon** : Reveals plain-text password for onboarding assistance.
- **Active / Inactive Toggle** : Instantly locks out staff account without deleting historical audit logs.
- **Edit Staff** : Updates user profile.
- **Delete** : Permanently deletes user account.
- **Tab 2 (Roles Matrix)**: Displays 15-module permission grid across all roles.

Path: /settings · Route: settings

Super Admin, Admin

Global enterprise profile, statutory registration details, printer hardware preferences, and database backup tools.

Company Profile Configuration

- **Trade Name & Legal Name** : Official business name printed on tax invoices.
- **Registered Address** : Factory and office postal address.
- **GSTIN & State Code** : Statutory 15-digit GSTIN (Tamil Nadu: 33).
- **FSSAI License** : Food safety registration number.
- **Bank Details** : Bank Name, Account Number, and IFSC Code printed on invoice footer.
- **Save Company Profile** : Commits updates to cloud and local storage.

Hardware & Printer Configuration

- **Default Print Format** : Switch between A4 Laser / Deskjet , 80mm Thermal , or 58mm Portable POS .
- **Auto-Print on Save** : Toggle to automatically trigger printer upon invoice creation.

Data Management

- **Export Database Backup (JSON)** : Full dump of customers, products, sales, and batches.
- **Reset Seed Data** : Factory reset option for demonstration environments.

Alphabetical index of all major interactive buttons across the application:

Button Label	Screen Location	Action / Behavior
+ Add Customer Account	/customers	Opens customer registration modal with credit limit and GSTIN validation.
+ Add New Employee	/employees	Creates staff account with username, password, and assigned role.
+ Add New Product Master	/products	Defines new SKU, tier rates (AWR), stock thresholds, and HSN code.
+ Add Supplier Vendor	/suppliers	Registers new vendor profile with materials, contact info, and payable balance.
+ Add to Voucher	/billing	Inserts product into active invoice table with automated tax calculations.
Adjust Physical Stock	/inventory	Opens dialog to reconcile physical stock count with recorded system inventory.
Cancel Bill	/billing-history	Prompts for cancellation reason, restores stock to godown, reverses ledger debit.
Clear Cart	/counter	Wipes all items from active retail POS cart.
Clear Form	/billing	Clears all customer, dispatch, and item fields for the next transaction.
Complete & Print Thermal	/counter	Finalizes POS sale, deducts inventory, and prints thermal receipt.
Export Master CSV	/products	Downloads complete product master catalog as spreadsheet.
+ Generate Dispatch Gate Pass	/logistics	Links vehicle, driver, and sales invoice to generate formal dispatch gate pass.

Button Label	Screen Location	Action / Behavior
Hold Cart	/counter	Temporarily parks active customer cart in local memory.
Print A4 Tax Invoice	/billing , /billing-history	Renders statutory A4 PDF invoice complete with bank details and signature blocks.
Print Thermal Receipt	/billing , /counter	Emits 80mm / 58mm ESC/POS slip to receipt printer.
+ Record Seed Inward	/purchase	Opens weighbridge seed procurement modal to record incoming raw material.
Record Milling Run	/production	Saves crushing batch, computes yield %, and adds finished oil to bulk tank.
Reprint A4	/billing-history	Re-opens A4 PDF preview dialog for historical invoice.
Reprint Thermal	/billing-history	Re-sends historical invoice copy to thermal printer.
Save Company Profile	/settings	Updates legal business name, address, GSTIN, FSSAI, and bank details.
Save Invoice	/billing	Validates form, creates sales order, deducts stock, updates customer credit ledger.
Sign In to ERP System	/login	Authenticates username and password, creates session, navigates to home route.
+ Start Milling Batch	/production	Opens production batch logging modal to record crushing run.

SOP 1: Cashier Shift Opening & POS Counter Billing

1. Sign in at `/login` using cashier credentials (`cashier`).
2. System opens `/counter` automatically. Count and verify cash drawer opening float.
3. For walk-in customers, tap product icons to populate cart. Adjust quantities using `+` / `-` steppers.
4. Select payment mode: `Cash` or `UPI QR` .
5. If cash, type customer amount into `Cash Received` ; confirm **Change to Return**.
6. Click **Complete & Print Thermal** . Hand receipt and oil to customer.
7. At end of shift, review `/billing-history` , verify cash against system totals, and hand over physical cash bag to Manager.

SOP 2: Creating a B2B Wholesale GST Tax Invoice

1. Navigate to `/billing` .
2. Select customer from `Customer Selector` . Confirm displayed credit balance.
3. Verify dispatch godown (`Main Godown`) and delivery route.
4. Select price tier: `Bulk Wholesale Rate` .
5. Search SKU (e.g. `PRD-GN-15L`), enter quantity (e.g. 50 tins), and click **+ Add to Voucher** .
6. Confirm automated CGST/SGST (intra-state) or IGST (inter-state) tax computation.
7. Click **Bottom Tab 0 (Dispatch)**; enter Vehicle Number, Driver Name, and Buyer PO Number.
8. If invoice exceeds ₹50,000, enter E-Way Bill Number in **Tab 1**.
9. Click **Save Invoice** , then click **Print A4 Tax Invoice** . Print original and transporter copies.

SOP 3: Mandi Seed Intake & Weighbridge Recording

1. Weigh loaded seed truck on weighbridge to obtain Gross Weight.
2. QC technician inspects seed sample for moisture content and kernel damage.
3. Navigate to `/purchase` and click **+ Record Seed Inward** .
4. Select vendor, enter material (`Groundnut Raw Pods`), net weight (Gross - Tare), and rate per kg.
5. Set QC Status: `Passed (Graded)` or `Pending Lab Test` .
6. Click **Save Inward Entry** to generate Goods Receipt Note (GRN) and update supplier payable ledger.

SOP 4: Expeller Milling Run & Yield Calculation

1. Allocate raw seeds (e.g. 5,000 kg groundnut pods) from warehouse to expeller batteries.
2. After crushing and filtration, measure recovered oil volume in settling tank.
3. Navigate to `/production` and click **+ Start New Production Milling Run** .
4. Enter Seed Input (5,000 kg), Filtered Oil Output (e.g. 2,150 Liters), and Bagged Cake Output (2,850 kg).
5. Verify extraction yield (40% - 43%) and click **Record Milling Run** . Bulk oil volume automatically updates in `/inventory` .

The mobile app is optimized for Android tablets and smartphones, empowering mill supervisors, drivers, and counter billers with full mobile capabilities.

Key Mobile Capabilities

- **5-Station Bottom Navigation:** Quick switching between Dashboard, Billing POS, Inventory, History, and the full ERP module drawer.
- **Force Desktop View Toggle:** Tap the topbar `Desktop View` toggle to view multi-column tables without mobile truncation.
- **Bluetooth Thermal Printing:** Pair with 58mm or 80mm wireless thermal printers for instant mobile receipt issuance.
- **Digital Proof of Delivery (POD):** Logistics drivers can record delivery completion in `/logistics` by tapping `POD Received` on mobile.